

Academic Career and Education

12/2024 – present **Research fellow in Computational Surface Chemistry**, University of Warwick, UK.

Supervisor: Reinhard Maurer

Postdoctoral role for developing embedding methods in the electronic structure code, FHI-aims.

08/2024 – 12/2024 **Research fellow in Computational Physical Chemistry**, University of Warwick, UK.

Supervisor: Gabriele Sosso

Short postdoctoral role (4 months) investigating ice propagation through plant cell walls using molecular dynamics.

01/2024 – 09/2024 **Freelance Software Developer**, Coventry, UK.

Contracted by Molecular Simulations from First Principles e.V. (MS1P) to develop the FHI-aims electronic structure code, specifically the density functional perturbation theory implementation. This work also includes the creation of tutorials for the FHI-aims summer school.

09/2021 – 01/2024 **Ph.D., Chemistry**, University of Warwick, UK.

07/2018 – 11/2020 Pass with minor corrections (Defended on 25/01/2024)

Including:

03/2023 – 01/2024 Employed as **Research assistant in computational chemistry**

Supervisor: Prof. Reinhard J. Maurer

Studied electronic friction as a method to simulate molecules on metal surfaces. Six publications produced, including one featured in a JACS Au special issue "Emerging Chemistry & Machine Learning", arising from international collaborations with China and Germany. Developer for electronic friction/electron-phonon coupling module within FHI-aims. > 15 successful applications for HPC time on various Tier 1 & 2 machines in the UK. Presented work in eight talks and 11 poster presentations given at conferences in UK, Europe, and China. Reviewed 1 paper for Physical Review B, co-reviewed 1 paper for Physical Review Letters. Co-organised weekly computational chemistry seminar at Warwick. Provided support for conferences/workshops at Warwick.

11/2020 – 09/2021 **Research fellow**, University of Warwick, UK.

Supervisors: Prof. Reinhard J. Maurer, Dr Andrew Logsdail, Dr Volker Blum, Dr Mariana Rossi, Dr Christian Carbogno

Technical staff for [ARCHER2-eCSE01-16](#) "Achieving the sustainability and scalability of numeric-atomic-orbital-based linear response and electron-phonon functionality in FHI-aims". Refactored density functional perturbation theory implementation within FHI-Aims. This work enabled new functionalities, modularized the code, reduced number of lines of code and resolved existing issues. Comprehensive documentation for both users and developers was also produced. Featured work showcased on the ARCHER2 website and Twitter.

10/2014 – 07/2018 **MChem, Chemistry**, University of York, UK.

First Class with Distinction

Publication record

9 publications. 4 first-author or joint first-author publications.

>110 citations with H-index = 6 ([Google Scholar](#) 30th May 2025)

[9] W. G. Stark, **C. L. Box**, ..., R. J. Maurer, arXiv preprint arXiv:2505.18147

[8] J. W. Abbot, ..., **C. L. Box**, ..., M. Rossi, arXiv preprint arXiv:2505.00125, 2025.

[7] **C. L. Box**, N. Hertl and R. J. Maurer, The Journal of Physical Chemistry Letters, 2024, **15** (51), 12520-12525.

[6] **C. L. Box**, W. G. Stark and R. J. Maurer, *Electronic Structure*, 2023, **5**, Publisher: IOP Publishing, 035005.

[5] Y. Zhang*, **C. L. Box***, T. Schäfer, A. Kandratsenka, A. M. Wodtke, R. J. Maurer and B. Jiang, *Physical Chemistry Chemical Physics*, 2022, **24**, Publisher: Royal Society of Chemistry, 19753–19760. (*These authors contributed equally)

- [4] Y. Litman, E. S. Pócs, **C. L. Box**, R. Martinazzo, R. J. Maurer and M. Rossi, *The Journal of Chemical Physics*, 2022, **156**, Publisher: American Institute of Physics, 194106.
- [3] Y. Litman, E. S. Pócs, **C. L. Box**, R. Martinazzo, R. J. Maurer and M. Rossi, *The Journal of Chemical Physics*, 2022, **156**, Publisher: American Institute of Physics, 194107.
- [2] O. A. Douglas-Gallardo, **C. L. Box** and R. J. Maurer, *Nanoscale*, 2021, **13**, Publisher: Royal Society of Chemistry, 11058–11068.
- [1] **C. L. Box***, Y. Zhang*, R. Yin, B. Jiang and R. J. Maurer, *JACS Au*, 2021, **1**, Publisher: American Chemical Society, 164–173. (*These authors contributed equally)

■ Funding/Award Record

- 2025 PSI-k collaborating early-career investigators (CECI) finalist (500 EUR)
- 2018 – present Successful HPC applications including MCC-HEC (EPSRC) and HPC-Midlands+ averaging >1M CPUh a year.
- 2018 - 2022 EPSRC PhD Studentship
- 2018 Received 50 % discount (400 EUR) on fee of the Sino-German Summer School "Hands-On DFT and Beyond" in Beijing, China.
- 2016 10-week summer research project funded by EPSRC (1 of 3 in the Department)

■ Academic Service

- 2024 Co-organisation of UK FHI-aims meeting, University of Warwick, UK.
- 2022 – present Peer review activity for Physical Review Letters (1) and Physical Review B (1).
- 2019 – 2022 Provided organisational support for 3 conferences in UK.
- 2018 – 2019 Co-organisation of the Warwick theory group departmental seminar.

■ Presentation Track Record

- 8 talks given at conferences in UK and Europe.
- 11 poster presentations given at conferences in UK, Europe and China.

Relevant talks:

- [4] **C. L. Box**, A. Logsdail, V. Blum, M. Rossi, C. Carbogno, and R. J. Maurer, "Current status of DFPT within FHI-aims", **Invited Talk**, Presented in part at the FHI-aims Developers' and Users' Meeting 2023, Hamburg, Germany, 2023.
- [3] **C. L. Box** and R. J. Maurer, "Quantifying the breakdown of electronic friction theory during molecular scattering of NO from Au(111)", **Contributed Talk**, Presented in part at the Royal Society meeting, "Software solutions to the challenges of materials modelling", Northampton, UK, 2022.
- [2] **C. L. Box** and R. J. Maurer, "Quantifying the breakdown of electronic friction theory during molecular scattering of NO from Au(111)", **Invited Seminar**, Presented in part at the Lorentz Center online seminar "Energy dissipation at interfaces", Online, 2021.
- [1] **C. L. Box** and R. J. Maurer, "Simulation of Electronic Friction Effects in Chemical Dynamics at Metal Surfaces: Understanding the Successes and Failures of Ab-initio Methods", **Contributed Talk**, Presented in part at the DPG spring meeting of the condensed matter section, University of Regensburg, Germany, 2019

Relevant Posters:

- [2] **C. L. Box**, N. Hertl, and R. J. Maurer, "Simulations of H atom scattering from Pt(111) surfaces with orbital-dependent friction", **Contributed Poster**, Presented in part at the 122nd Annual Conference of the German Bunsen Society for Physical Chemistry, Berlin, Germany, 2023.
- [1] **C. L. Box** and R. J. Maurer, "Quantifying the breakdown of electronic friction theory during molecular scattering of NO from Au(111)", **Contributed Poster**, Presented in part at the Psi-k community 2022 conference, Lausanne, Switzerland, 2022.

■ Teaching Experience

- 2018 – 2020 **Sessional teacher**, University of Warwick, UK.
- Taught workshops and implemented infrastructure for automated creation, distribution, submission, and grading of python-based assignments using Jupyter notebooks.
 - Provided guidance and support to academics at Warwick University and Nottingham Trent University on deployment, including live demonstrations and documentation creation.

■ Membership of Professional Societies

- 2018 - present **Institute of Physics**
- 2011 – present **Royal Society of Chemistry**